Some of my white boardwriting from week 10

Some students insisted that I put in writing my calculations on the white boards in relations to the fitness data. I should stress that I already provided an annotated output for the chicken data. The calculations for the fitness data are along same lines. I strongly urge you to go through the lecture notes first before looking how to do particular calculations. Anyhow, below are the main explanations/calculations related to the fitness data:

We have $p = q = 3 = \nu_h$ since both physical and exercise variables are 3-dimensional here.

I pointed out that always $\nu_h + \nu_e$ = Sample size -1 (since we always "lose" one df when estimating the general mean; i.e. in the decomposition $SST = SSA + SSE$, the df on left are $(n-1)$).

Therefore $\nu_e = 20 - 1 - 3 = 16 = \nu$. Based on these we calculate the triplet (s, m, n) (denoted as capital S, M, N in the SAS output): $S = \min(p, q) = 3$,

$M = \frac{1}{2}\left(|p-q| - 1\right) = -\frac{1}{2}$, $N = \frac{1}{2}(\nu_e - p - 1) = \frac{16-3-1}{2} = 6$

In the context of canonical correlations:

$H = \Sigma_{21}\Sigma_{11}^{-1}S_{12} = Q_2$ (the hypothesis matrix)

$E = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}S_{12} = Q_1$ (the error matrix)

The eigenvalues λ_i I refer to in the notes, are the eigenvalues of $E^{-1}H = Q_1^{-1}Q_2$.

However, they can also be expressed with the eigenvalues of $\Sigma_{22}^{-1}H$ (and those we denoted (see notes) as $\mu_1^2, \mu_2^2, \mu_3^3$).

From $|E^{-1}H - \lambda I| = 0$ we get easily

$$\left|\frac{1+\lambda}{\lambda}H - \Sigma_{22}\right| = 0 \rightarrow \left|\Sigma_{22}^{-1}H - \frac{\lambda}{1+\lambda}I\right| = 0$$ which

implies that $\mu_i^2 = \frac{\lambda_i}{1+\lambda_i}$, $i=1,2,3$ must hold.

(or equivalently $\lambda_i = \frac{\mu_i^2}{1-\mu_i^2}$ $i=1,2,3$ must hold).

The Wilks' statistic is defined (see notes) as

$\Lambda = |Q_1(Q+Q_2)^{-1}| = |E(E+H)^{-1}|$. For $E^{-1}H$, the eigenvalues: $\lambda_1 = 1.7247$, $\lambda_2 = 0.0419$, $\lambda_3 = 0.0053$ are listed in the SAS output. Lemma 2 of the lecture notes gives the eigenvalues for Wilks as

$\frac{1}{1+\lambda_1}, \frac{1}{1+\lambda_2}, \frac{1}{1+\lambda_3}$ and the Wilks criterion itself

is a <u>determinant</u>, i.e., product of the eigenvalues,

So $\Lambda = \frac{1}{1+1.7247} * \frac{1}{1+0.0419} * \frac{1}{1+0.0053} = .35039053$

Pillai's trace is the <u>sum</u> (see notes)

$$\frac{\lambda_1}{1+\lambda_1} + \frac{\lambda_2}{1+\lambda_2} + \frac{\lambda_3}{1+\lambda_3} = \frac{1.7247}{1+1.7247} + \dots = .67848157$$

Hotelling-Lawley is just the sum

$\lambda_1 + \lambda_2 + \lambda_3 = 1.7247 + 0.0419 + 0.0053 = 1.77194146$

and Roy's greatest root is just $\lambda_1 = 1.7247$

The df for the corresponding F-approximations of these root-based statistics are also explained in the notes. Here I just plug-in to explain how

they are calculated for the Wilk's statistic:
Based on the triplet $s=3$, $M=-\frac{1}{2}$, $N=6$ we go on as follows:

- Num DF $= pq = 3*3 = 9$ (easy)
- Den DF:

$$r = \nu_e - \frac{1}{2}(p-q+1) = 16 - \frac{1}{2}(3-3+1) = 15.5$$

$$t = \sqrt{\frac{81-4}{18-5}} = \sqrt{\frac{77}{13}} = 2.434$$

$$u = \frac{1}{4}(pq-2) = 7/4$$

$$\text{Den DF} = rt - 2u = 15.5 * 2.434 - 2*\frac{7}{4} = \underline{34.223}$$

The test of H_0: The canonical correlations in the current and subsequent rows are zero is based on the Λ statistic so Lik Ratio $= .35039053$ p-value $= .0635$
The remaining statistics in this table indicate that the second and third canonical correlations are indeed meagre and can be disregarded in terms of any importance.

<u>I ALSO</u> showed why Wilks' Λ is indeed just $\Lambda = \frac{|S|}{|S_{11}||S_{22}|}$

This is because

$$\Lambda = |E(E+H)^{-1}| = |(S_{22} - S_{21}S_{11}^{-1}S_{12})S_{22}^{-1}| =$$

$$= |\underbrace{S_{11}(S_{22} - S_{21}S_{11}^{-1}S_{12})}_{\text{now } \det(\dots) = \det S \text{ as discussed in lectures}} S_{22}^{-1}S_{11}^{-1}| = |S||S_{22}^{-1}||S_{11}^{-1}|$$

$$= \frac{|S|}{|S_{11}||S_{22}|}$$

<u>I ALSO Discussed thoroughly the</u>. <u>whole</u> <u>SAS output of the fitness data</u> but you can find this discussion in the <u>help of the cancorr procedure</u> so I abstain from reproducing it here